



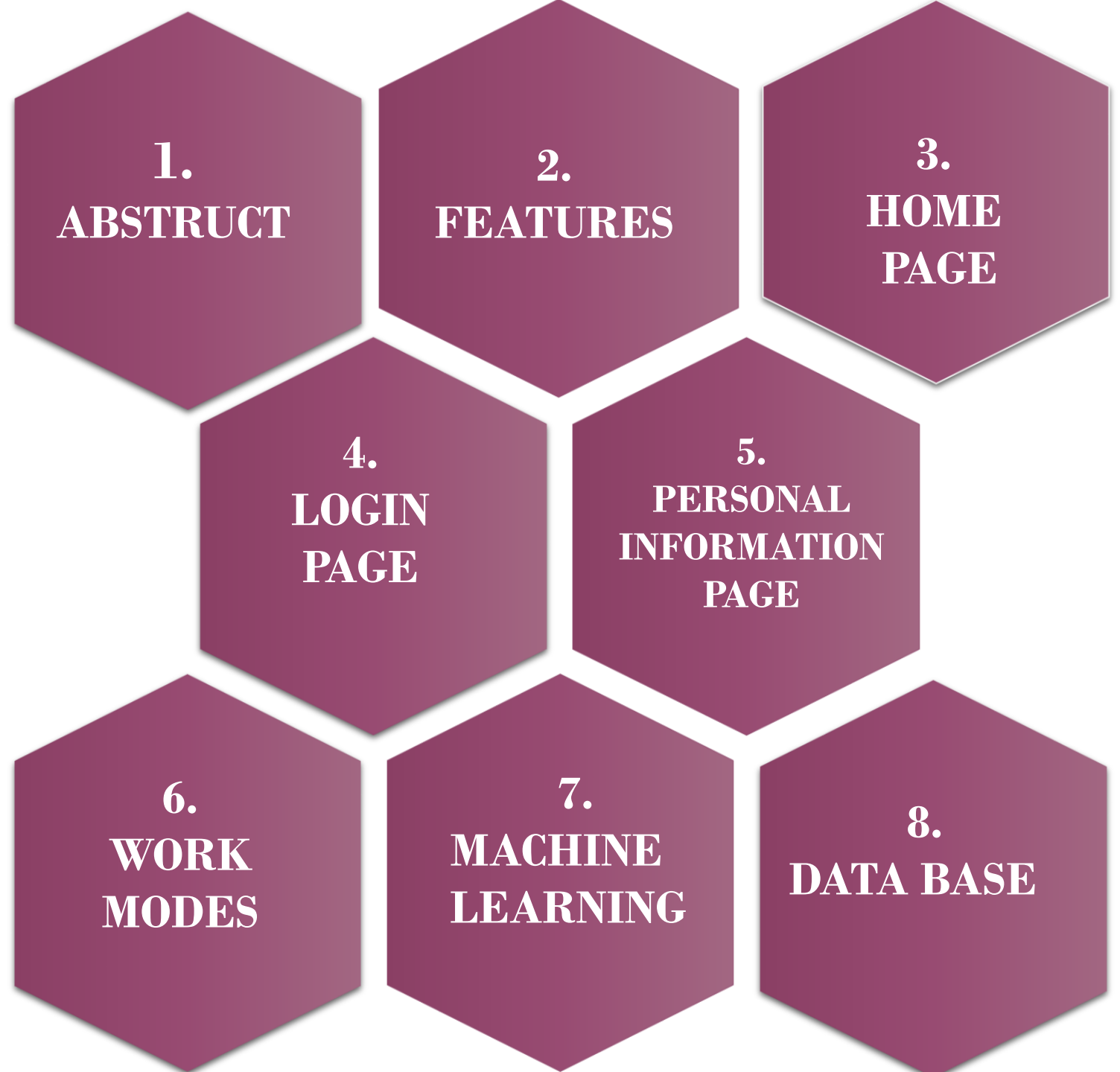
WELCOME

TO OUR PRESENTATION

CREDIT CARD APPROVALS SYSTEM USING MACHINE LEARNING

- JAGADEESWARY.T

MAIN MODULES



TOOL'S AND TECHNIQUES:

Are used

SOFTWARE REQUIREMENTS

- Frontend : HTML , JAVASCRIPT
- Backend : MySQL

HARDWARE REQUIREMENTS

- OS: Windows10
- Hard Disk: 1TB
- RAM: 4GB (Recommended)

FLOW DIAGRAM

User fills form



- **Personal Info + Work Details collected**



- **Neural Network trains**



- **Applicant's data fed into trained model**



- **Model outputs probability score (0–1)**



- **Score + hard rules → Final Decision**



- **Approved → Show recommended banks**

- **Not Eligible → Show alternative banks + improvement tips**

ABOUT OUR PROJECT:

-Abstract

- Nowadays credit card have been increasing exponentially.
- Art of machine learning is that it can be used to solve various problems given the historical data.
- In this blog, we will attempt to predict how likely a credit card request will approve.
- The project focus on visualizing and analyzing data, prepossessing, model building and prediction.



FEATURES:



ML powered
decision



Real Time
Processing



Secure and
Private



Bias
Detection



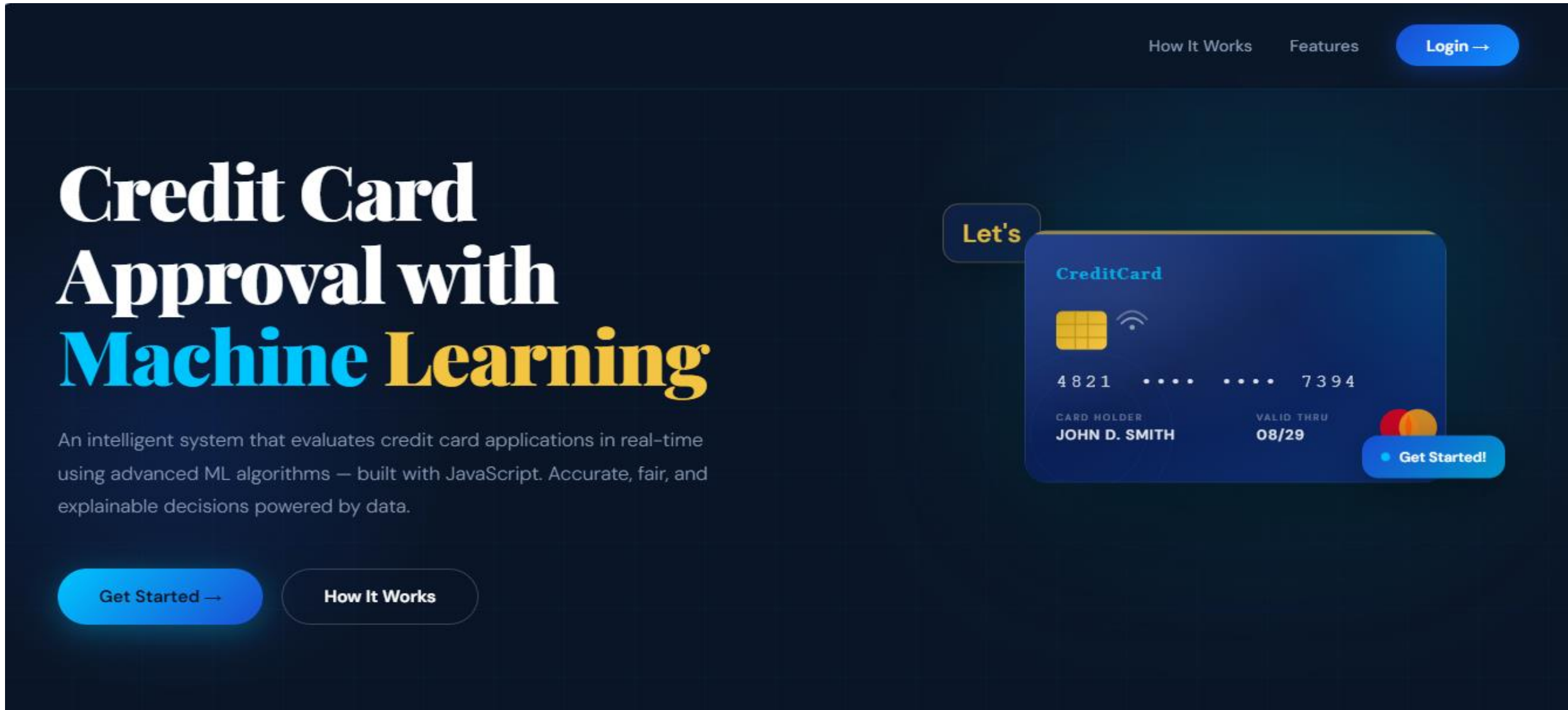
Analytics
Dashboard

HOME PAGE:

- ❖ The Home page contains how the system works , features and login button.
- ❖ **System works** explain the process from raw application data into a final credit decision.
- ❖ **Key features** explain the technical and usage of this project.



SCREEN SHOT: - Home page



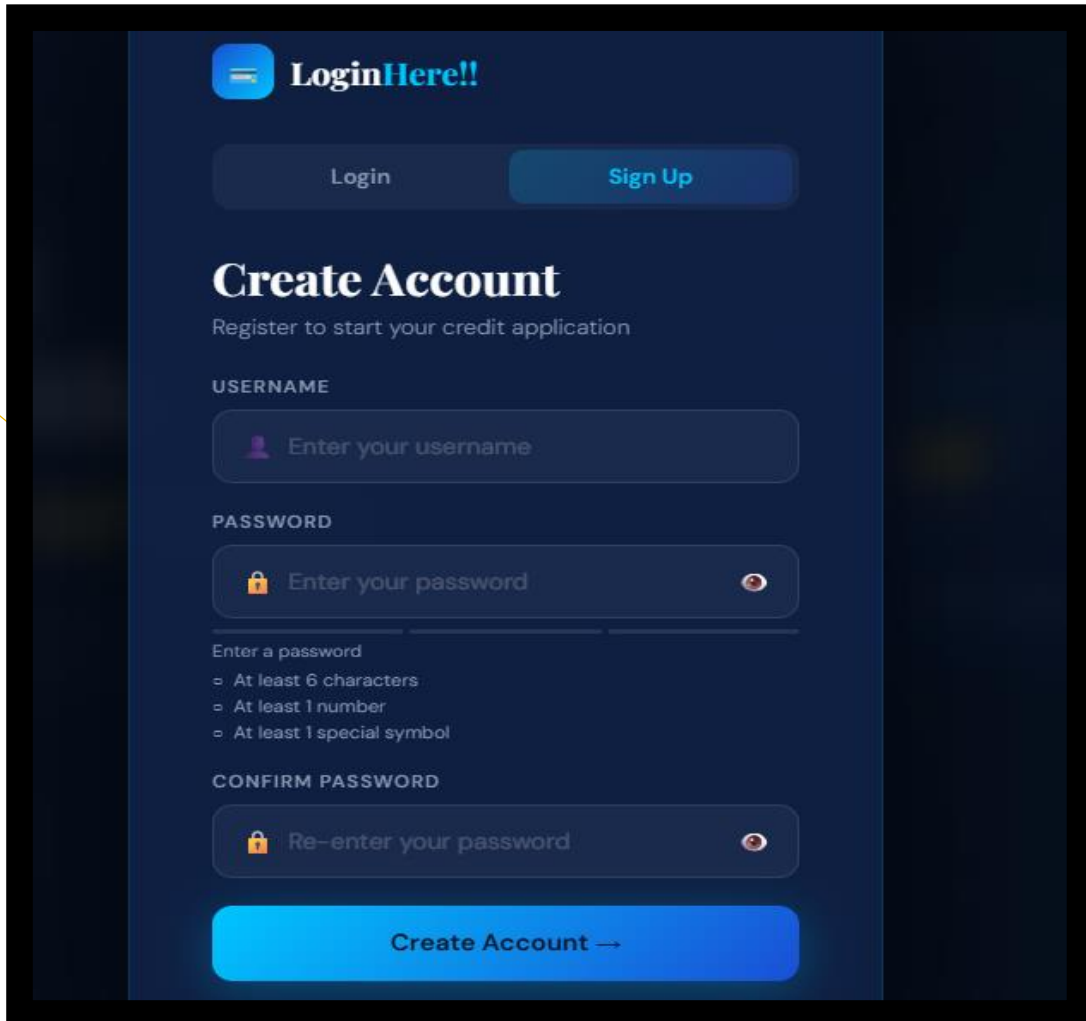


LOGIN PAGE:

- The login page contains two modules: **Sign Up** and **Login**.
- A new user must go to the **Sign Up** page and fill in the required details to create an account.
- After signing up, the user can continue the process by logging in using their credentials.

SCREEN SHOTS: - LOGIN PAGE

NEW USER



The screenshot shows a dark-themed 'Create Account' form. At the top left is the 'LoginHere!!' logo. Below it are 'Login' and 'Sign Up' buttons. The main heading is 'Create Account' with the subtext 'Register to start your credit application'. The form includes three input fields: 'USERNAME' (with a person icon), 'PASSWORD' (with a lock icon and a toggle for visibility), and 'CONFIRM PASSWORD' (with a lock icon and a toggle for visibility). Below the password fields, there is a list of password requirements: 'Enter a password', 'At least 6 characters', 'At least 1 number', and 'At least 1 special symbol'. At the bottom is a large blue 'Create Account →' button.

LoginHere!!

Login Sign Up

Create Account

Register to start your credit application

USERNAME

Enter your username

PASSWORD

Enter your password

Enter a password

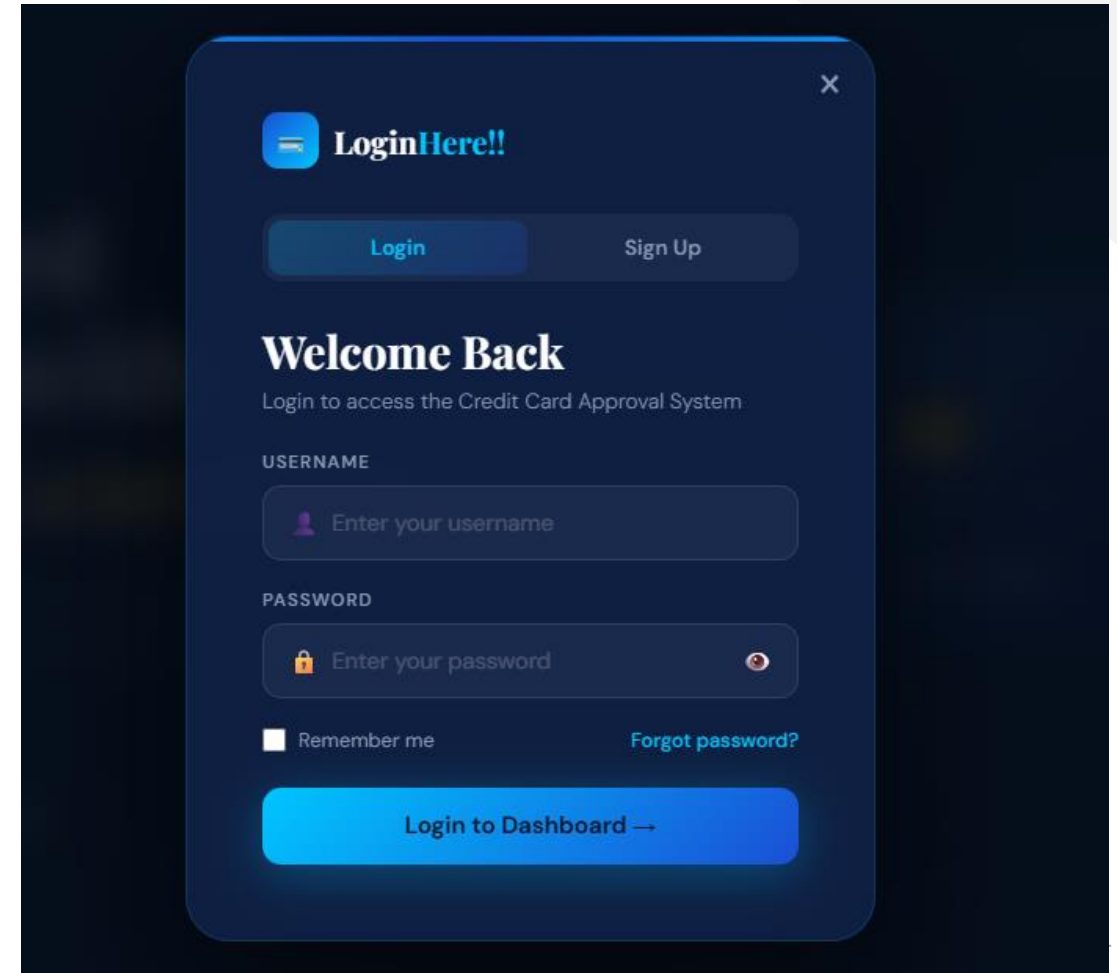
- At least 6 characters
- At least 1 number
- At least 1 special symbol

CONFIRM PASSWORD

Re-enter your password

Create Account →

SIGNUP



The screenshot shows a dark-themed 'Welcome Back' login form. At the top left is the 'LoginHere!!' logo with a close button (X) in the top right corner. Below it are 'Login' and 'Sign Up' buttons. The main heading is 'Welcome Back' with the subtext 'Login to access the Credit Card Approval System'. The form includes two input fields: 'USERNAME' (with a person icon) and 'PASSWORD' (with a lock icon and a toggle for visibility). Below the password field are checkboxes for 'Remember me' and a link for 'Forgot password?'. At the bottom is a large blue 'Login to Dashboard →' button.

LoginHere!!

Login Sign Up

Welcome Back

Login to access the Credit Card Approval System

USERNAME

Enter your username

PASSWORD

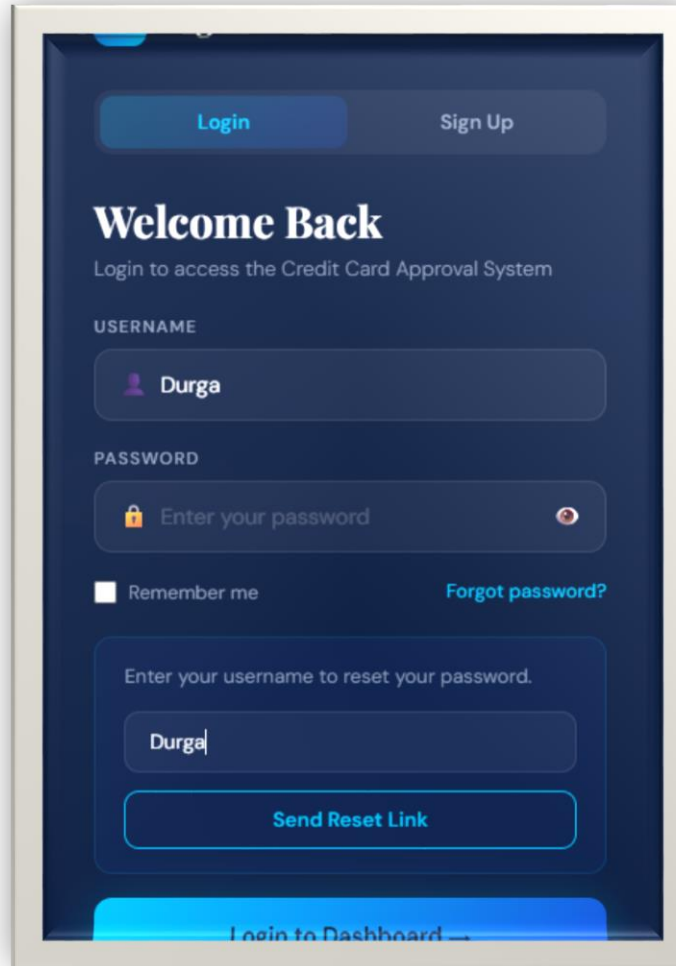
Enter your password

☐ Remember me [Forgot password?](#)

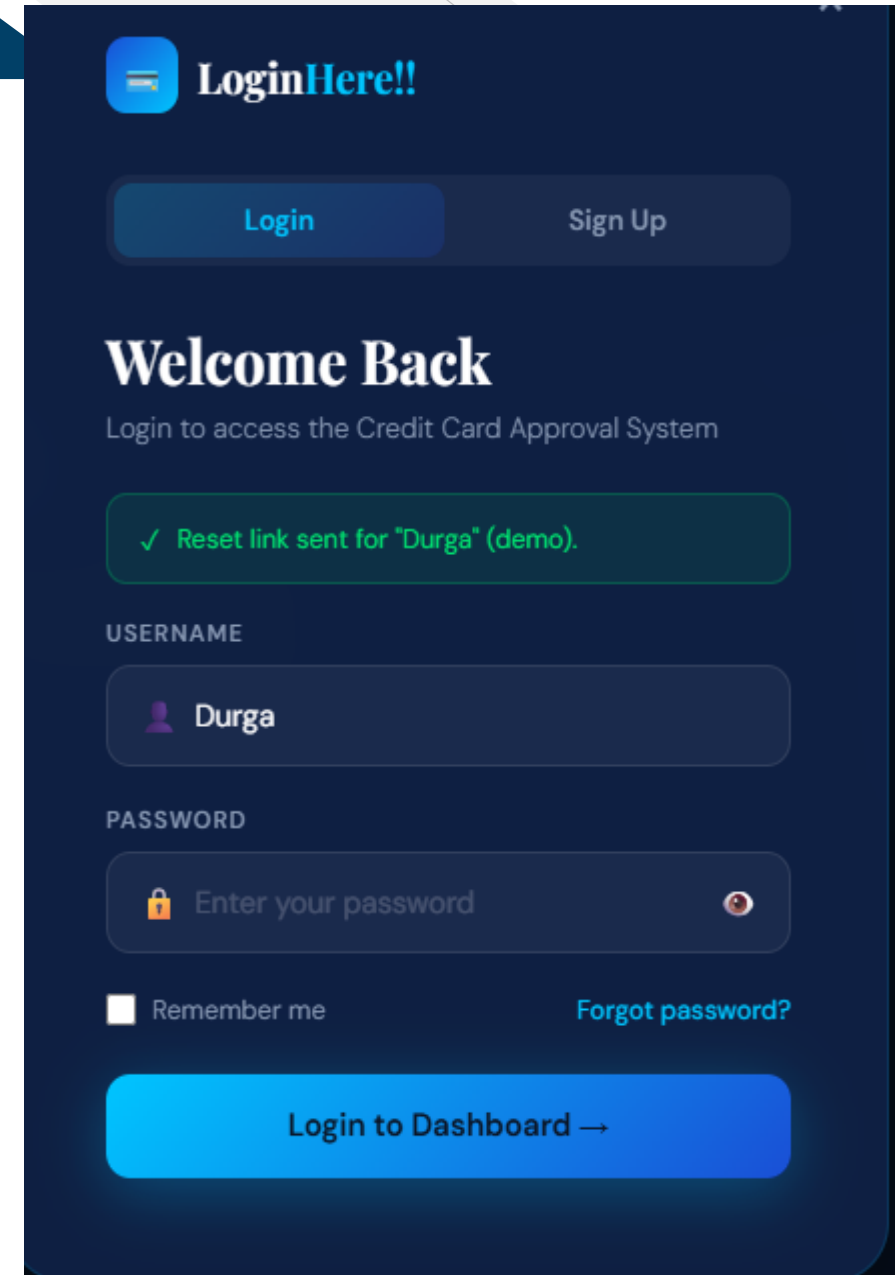
Login to Dashboard →

FORGOT-PASSWORD

System checks
the valid
username given
by the user then
provide the reset
portal



This screenshot shows a 'Forgot Password' portal. At the top, there are 'Login' and 'Sign Up' buttons. The main heading is 'Welcome Back' with the subtitle 'Login to access the Credit Card Approval System'. Below this, there is a 'USERNAME' field containing 'Durga'. A 'PASSWORD' field is present with a lock icon and a toggle for visibility. There is a 'Remember me' checkbox and a 'Forgot password?' link. A section below the password field asks the user to 'Enter your username to reset your password.' with 'Durga' entered in the field. A 'Send Reset Link' button is at the bottom of this section. At the very bottom, a 'Login to Dashboard' button is partially visible.



This screenshot shows a 'Login Here!!' portal. At the top, there is a logo and the text 'LoginHere!!'. Below this are 'Login' and 'Sign Up' buttons. The main heading is 'Welcome Back' with the subtitle 'Login to access the Credit Card Approval System'. A green success message states '✓ Reset link sent for "Durga" (demo)'. Below this, there is a 'USERNAME' field containing 'Durga'. A 'PASSWORD' field is present with a lock icon and a toggle for visibility. There is a 'Remember me' checkbox and a 'Forgot password?' link. A large 'Login to Dashboard →' button is at the bottom.

PERSONAL INFORMATIONS:

- ❑ In the credit card approval process, personal information is collected to understand the applicant and decide whether to approve the credit card
- ❑ Basic details such as name, age, address are gathered to verify identity and check eligibility.
- ❑ This helps collecting basic details about a person to understand who they are and how they can be contacted and reduce the risk of fraud.



SCREEN SHOT: - Personal Information

Personal Information

Step 2 of 3 — Complete your profile to proceed with the credit card application

PERSONAL DETAILS

FULL NAME *

Enter full name

CONTACT / MOBILE *

Mobile number

GENDER *

Select Gender

DATE OF BIRTH *

dd-mm-yyyy

AGE (AUTO)

Auto-calculated

EMAIL ADDRESS *

your@email.com

ADDRESS

HOUSE NO. *

e.g. 42B

STREET *

Street name

VILLAGE / TOWN *

Village or town

DISTRICT *

District

Continue to Work Details →

WORK MODE

The work mode is divided into
Two:



**OWN BUSINESS
SECTOR**



Work Mode

Step 3 of 3 – Select the category that best describes your current employment or business status.

Own Business
Self-employed or running your own venture →

Sector
Government or private sector employment →

OWN BUSINESS

- The process involves collecting specific information about a user's business to assess credit card eligibility using a machine-learning (ML) model.
- 1. *Business Information Collected*:
 - - *Business Name*: The user's company name (e.g., Textile Trading, Bakery, IT Consulting).
 - - *Business Type*: Category of the business (e.g., Manufacturing, Services, Trading).
 - - *Years in Business*: Number of years the business has been operational.
 - - *Yearly Income*: Annual earnings of the business in ₹ (e.g., ₹8,00,000).
- 2. *Purpose of Data Collection*:
 - - To evaluate the business's financial stability and creditworthiness.
 - - To feed these features into an ML model that predicts credit card approval probability.

SCREEN SHOTS: -Own Business

[← Back](#)



Your Business

Tell us about your business and annual earnings for a complete credit assessment.

BUSINESS NAME *

BUSINESS TYPE *

Select type

YEARS IN BUSINESS *

BUSINESS TYPE *

Select type

YEARS IN BUSINESS *

YEARLY INCOME (₹) *

₹ e.g. 800000

Submit Business Details →

SECTORS

PRIVATE SECTOR

- **Business Name** – The name of the company, organization, or business where the person works.
- **Experience** – The total number of years the person has worked in that private company or in the private sector.
- **Yearly Income** – The total amount of money the person earns in one year from the private job or business.

GOVERNMENT SECTOR

- **Department** – The name of the government department or office where the person works.
- **Experience** – The number of years the person has worked in the government sector.
- **Yearly Income** – The total salary the person receives in one year from the government job.
- **Permanent or Temporary** – Indicates whether the government job is **permanent (long-term position)** or **temporary (contract or limited-time position)**.

SCREEN SHOTS

GOVERNMENT WORK

Government Work

Provide your government employment details for a complete credit assessment.

DEPARTMENT / MINISTRY *

e.g. Railways, Health Ministry, PWD, ISRO

DESIGNATION / ROLE *

e.g. Junior Engineer, Clerk, IAS Officer

EMPLOYMENT TYPE *

☒ Permanent ☐ Temporary

YEARS OF SERVICE *

e.g. 8

YEARLY ANNUAL INCOME (₹) *

₹ e.g. 720000

PRIVATE WORK

Private Work

Tell us about your private sector employment for a complete credit profile.

COMPANY NAME *

e.g. Infosys, Reliance, Zomato

INDUSTRY / SECTOR *

Select industry ▼

ROLE / DESIGNATION *

e.g. Software Engineer, Sales Manager

EXPERIENCE (YEARS) *

e.g. 4

YEARLY ANNUAL INCOME (₹) *

₹ e.g. 1200000

APPLICATION COMPLETE!

GOVERNMENT

PRIVATE

WORK SUMMARY

Work Type	Government Work
Employee In	Educational
Designation	Teacher
Employment Type	Permanent
Years of Service	3 years
Yearly Income	₹ 4,00,000

 [Check Credit Card Eligibility →](#)

[← Start Over](#)



Application Complete!

Your personal and work information has been recorded successfully. Your credit card application is now under review.

WORK SUMMARY

Work Type	Private Work
Company	Cognizant
Industry	Information Technology
Role	Software Engineering
Experience	1 years
Yearly Income	₹ 3,00,000

CHECK ELIGIBILITY

Our credit card approval system checks two main criteria:

- 1. **Age**: The applicant must be above 18 years old.
- 2. **Annual Income**: The applicant's annual income must be ₹75,000 or more.

Approval Process:

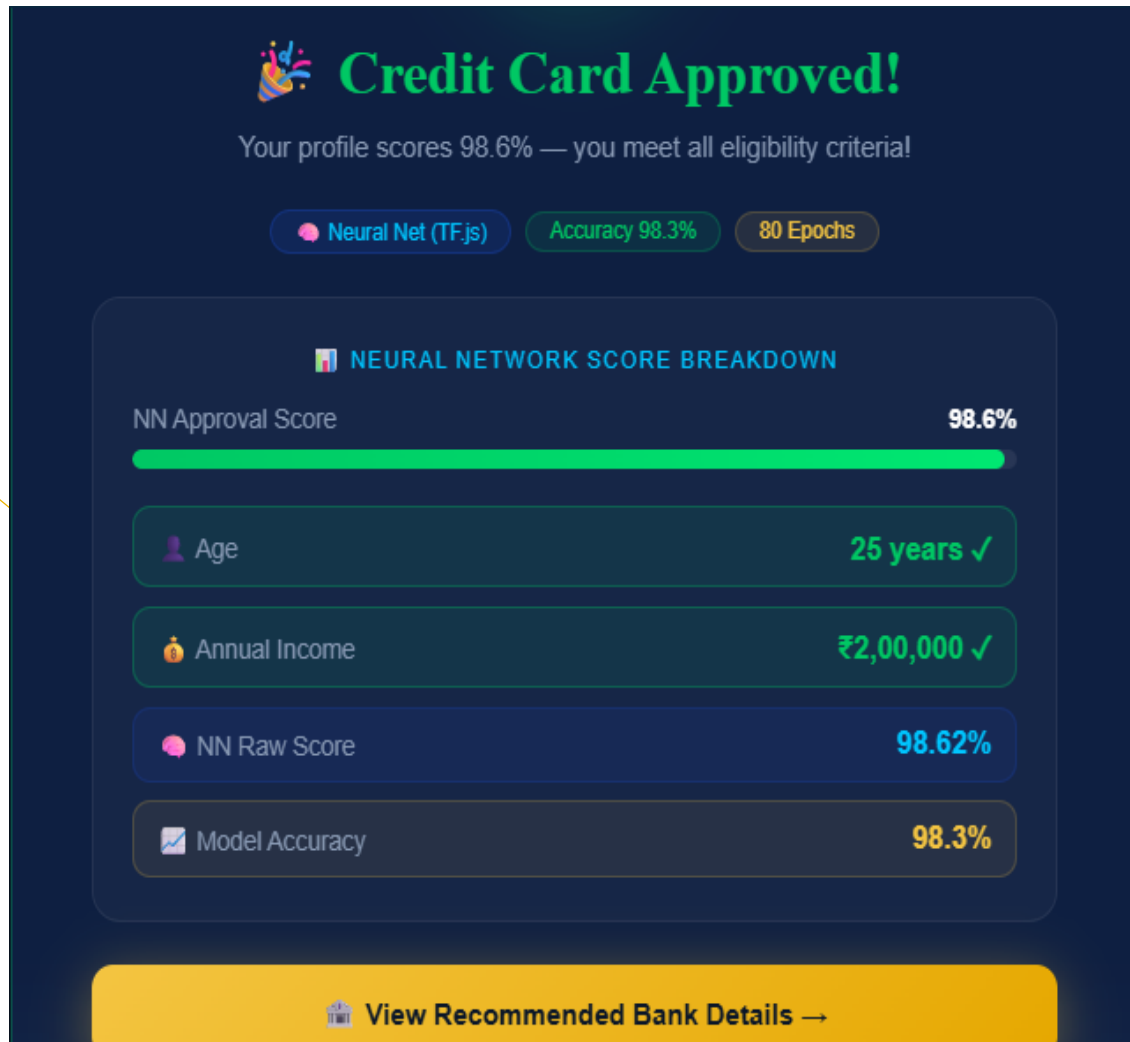
- - If the applicant meets both criteria, their application is approved.
- - If the applicant does not meet either of the criteria, their application is rejected.

Example Scenarios

- - **Approved**: Applicant is 25 years old with an annual income of ₹1,00,000.
- - **Rejected**: Applicant is 17 years old with an annual income of ₹50,000.

SCREEN SHOTS:

APPROVED



REJECTED



SUGGESTION

- We give the GOVERNMENT BANKS Suggestion for the persons who are eligible .The bank details under eligibility , Required Documents to get Credit Card and EMI details.
- For a non-eligible persons we display the reason for non-eligibility and show some private banks details like eligibility ,EMI , Required documents

(Note - “ All the bank details are gathered from regarding banks portal. ”)



BANK DETAILS

FOR ELIGIBLE PERSONS

[← Back to Result](#)



Recommended Banks

Age 25 (21+) — Full-service bank options



1. State Bank of India

✓ Eligible

ELIGIBILITY

- Age: 21–70 years
- Occupation: Salaried or Self-Employed
- Regular source of income
- Good credit score required

EMI DETAILS

- Min Transaction: ₹500 (merchant) / ₹2,500 (Flexipay)
- Tenure: 3, 6, 9, 12, 18, 24, 36 months
- Interest: ~12%–24% p.a.
- Processing Fee: 2% (min ₹99, max ₹1,000)

DOCUMENTS REQUIRED

PAN CardAadhaar Card (first 8 digits masked)Salary Slip / ITRValid Govt Address Proof



2. Bank of Baroda

✓ Eligible

ELIGIBILITY

- Age: 21–65 years
- Residential Status: Indian Resident
- Employment: Salaried or Self-Employed
- Income: Varies by card variant

EMI DETAILS

- Min Transaction: ₹2,500
- Tenure: 6, 12, 18, 24, 36 months
- Interest Rate: 13%–15% p.a.
- Processing Fee: 2% (min ₹100)

DOCUMENTS REQUIRED

PAN CardAadhaar CardSalary Slip / ITRAddress Proof



3. Punjab National Bank

✓ Eligible

ELIGIBILITY

- Age: 21–70 years (add-on: min 18)
- Must be literate
- Independent financial means
- Net Annual Income: ₹2.5L–₹7.5L (by card type)

EMI DETAILS

- Tenure: 3, 6, 9, 12 months
- Interest Rate: ~13%–16% p.a.
- Processing Fee applicable
- Min income: ₹250,000 p.a. (Classic)

DOCUMENTS REQUIRED

PAN CardAadhaar CardITR / Salary SlipAddress Proof

FOR NON-ELIGIBLE PERSONS

[← Back to Result](#)



Alternative Bank Options

Although you don't currently qualify, these **5 private banks** offer flexible eligibility — work on your profile and apply soon.



How to improve your eligibility

- Maintain a **CIBIL score of 750+** by paying bills on time
- Ensure your **annual income exceeds ₹75,000**
- Stay employed continuously for **6–12 months**
- Start with a **secured/prepaid card** to build credit history

TOP 5 PRIVATE BANKS — ELIGIBILITY & EMI DETAILS

#	BANK	ELIGIBILITY	EMI MIN & TENURE	RATE	FEE
1	HDFC Bank Private Sector Bank	- Age: 21–60 yrs (sal) / 21–65 yrs (self-emp) - Income: >₹20,000/mo ITR >₹6L p.a. - CIBIL Score: 700+ - Indian Resident	- Min Transaction: ₹5,000 - Tenure: 3, 6, 9, 12, 18, 24 months	16% p.a.	₹199 + GST
2	ICICI Bank Private Sector Bank	- Age: 21–65 yrs - Income: ₹20,000/mo (sal) ₹3L p.a. (self-emp) - CIBIL Score: 750+ - Indian Citizen / Resident	- Min Transaction: ₹3,000 - Tenure: 3, 6, 9, 12, 18, 24 months	3.75%/mo (45% p.a.)	2% of transaction
3	Axis Bank Private Sector Bank	- Age: 18–70 yrs - Income: ₹25,000/mo (existing) ₹50,000/mo (new) - CIBIL Score: 750+ - Indian Resident or NRI	- Min Transaction: ₹2,500 - Tenure: 6, 9, 12, 18, 24, 36, 48 months	3.4%/mo (~40.8% p.a.)	As per card variant
4	Kotak Mahindra Bank Private Sector Bank	- Age: 21+ yrs - Income: ₹25,000/mo (₹3L p.a. min) - CIBIL Score: 750+ - Indian Resident	- Min Transaction: ₹2,500 - Tenure: 3 to 48 months - No-cost EMI on select brands	Reducing balance	N/A

MACHINE LEARNING

Powered by "TensorFlow.js Neural Network".

ML Model:

Type: Feedforward Neural Network

Framework: TensorFlow.js v4.17.0

Task: Binary Classification
(Approved / Not Eligible)

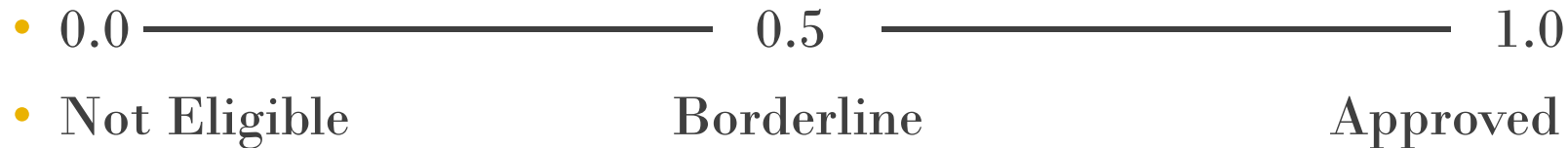
“Runs entirely in the browser — no server needed”.



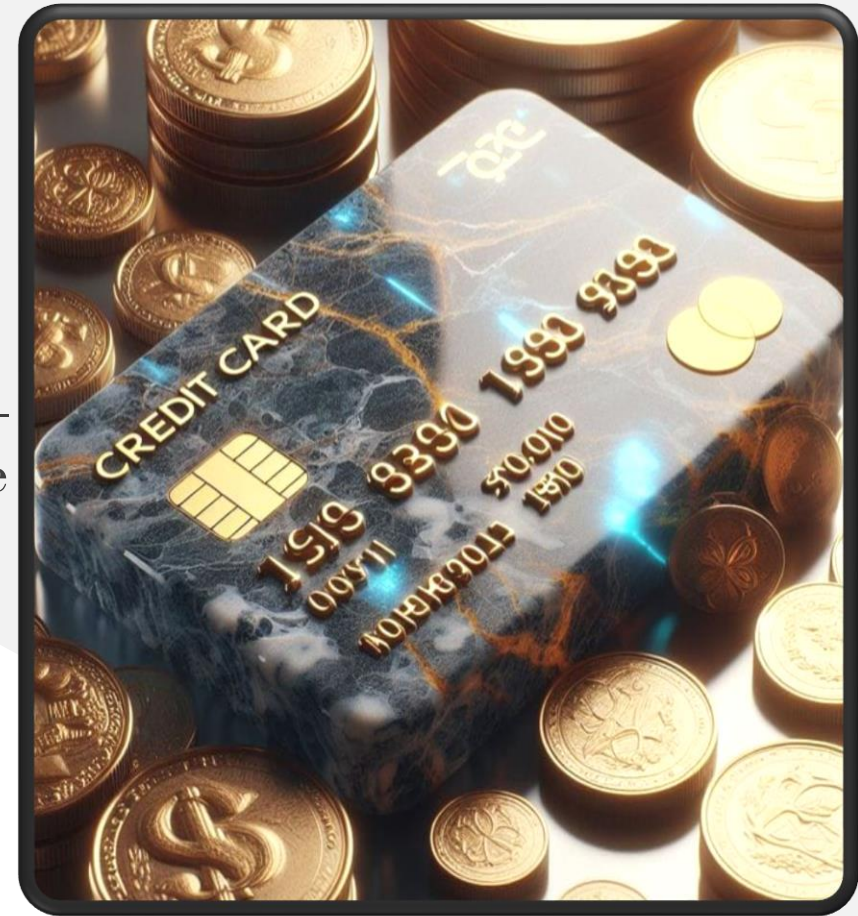
NEURAL NETWORK

Neural Network Used For:

- To predict whether a person should get a credit card or not.
- It reads a person's details — age, income, employment years — and outputs a single number between 0 and 1 that represents the probability of approval.



- **? WHY Neural Network**
- A simple rule like:
- if $\text{age} \geq 18$ AND $\text{income} \geq 75000 \rightarrow \text{Approve}$
- else $\rightarrow \text{Reject}$



DATABASE SCREEN SHOT:

phpMyAdmin

Server: 127.0.0.1 » Database: creditai_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Mon

Recent Favorites

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ activity_log	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ card_applications	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ credit_details	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ employment_info	★ Browse Structure Search Insert Empty Drop	13	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ ml_snapshots	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ password_resets	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_unicode_ci	48.0 KiB	-
☐ personal_info	★ Browse Structure Search Insert Empty Drop	17	InnoDB	utf8mb4_unicode_ci	32.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	16	InnoDB	utf8mb4_unicode_ci	64.0 KiB	-
☐ vw_application_summary	★ Browse Structure Search Insert Edit Drop	~0	View	---	-	-
☐ vw_approval_rate	★ Browse Structure Search Insert Edit Drop	~0	View	---	-	-
10 tables Sum		~53	InnoDB	utf8mb4_unicode_ci	368.0 KiB	0 B

☐ Check all With selected:

Print Data dictionary Console



Recent Favorites

- New
- credital_db
 - Procedures
 - Tables
 - New
 - activity_log
 - card_applications
 - credit_details
 - employment_info
 - ml_snapshots
 - password_resets
 - personal_info
 - users
 - Views
- information_schema
- mysql
- performance_schema
- phpmyadmin
- test

[Table structure](#)
[Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 log_id	bigint(20)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2 user_id	int(10)		UNSIGNED	Yes	NULL			Change Drop More
☐	3 action	varchar(120)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4 ip_address	varchar(45)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	5 user_agent	varchar(255)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	6 logged_at	datetime			No	current_timestamp()			Change Drop More

☐ Check all
 With selected:
 [Browse](#)
[Change](#)
[Drop](#)
[Primary](#)
[Unique](#)
[Index](#)
[Spatial](#)
[Fulltext](#)

[Add to central columns](#)
[Remove from central columns](#)

[Print](#)
[Propose table structure](#)
[Track table](#)
[Move columns](#)
[Normalize](#)

Add 1 column(s) after logged_at [Go](#)

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	log_id	0	A	No	
Edit Rename Drop	idx_user	BTREE	No	No	user_id	0	A	Yes	
Edit Rename Drop	idx_action	BTREE	No	No	action	0	A	No	
Edit Rename Drop	idx_time	BTREE	No	No	logged_at	0	A	No	



Recent Favorites

New

creditai_db

Procedures

Tables

New

activity_log

card_applications

credit_details

employment_info

ml_snapshots

password_resets

personal_info

users

Views

information_schema

mysql

performance_schema

phpmyadmin

test

Server: 127.0.0.1 » Database: creditai_db » Table: personal_info

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

Triggers

Table structure

Relation view

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	info_id	int(10)		UNSIGNED	No	None		AUTO_INCREMENT	Change Drop More
☐	2	user_id	int(10)		UNSIGNED	No	None			Change Drop More
☐	3	full_name	varchar(120)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	4	dob	date			No	None			Change Drop More
☐	5	gender	enum('male', 'female', 'other')	utf8mb4_unicode_ci		No	None			Change Drop More
☐	6	phone	varchar(15)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	7	email	varchar(120)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	8	address	text	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	9	city	varchar(80)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	10	state	varchar(80)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	11	pincode	varchar(10)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	12	pan_number	varchar(10)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	13	aadhar_last4	char(4)	utf8mb4_unicode_ci		Yes	NULL			Change Drop More
☐	14	created_at	datetime			No	current_timestamp()			Change Drop More
☐	15	updated_at	datetime			Yes	NULL		ON UPDATE CURRENT_TIMESTAMP()	Change Drop More



THANK YOU!

DATA ANALYTICS PROJECT

Electric Vehicle Population Analysis

Analyzing 150,000+ U.S. EV registration records to uncover adoption trends, manufacturer market share, and eligibility insights using Power BI & SQL

Jagadeeswary T

Data Analyst | jagadeeswary2006@gmail.com | [linkedin.com/in/jagadeeswary-t0906](https://www.linkedin.com/in/jagadeeswary-t0906)

Project Objective & KPI Requirements

01

Total Vehicles

Assess the overall size and growth of the EV market across both BEVs and PHEVs.

02

Average Electric Range

Gauge the technological advancement and efficiency of vehicles in the dataset.

03

BEV Share

Quantify Battery Electric Vehicles and their share of the total EV population.

04

PHEV Share

Quantify Plug-in Hybrid Electric Vehicles and their share of the market.

Visualization Approach & Tools

Line / Area Chart

Vehicles by Model Year

Growth pattern & adoption trend from 2010 onward

Map Chart

Vehicles by State

Geographic distribution and regional adoption hotspots

Bar Chart

Top 10 by Manufacturer

Market dominance of leading EV brands

Donut Chart

CAFV Eligibility

Share of vehicles eligible for clean-fuel incentives

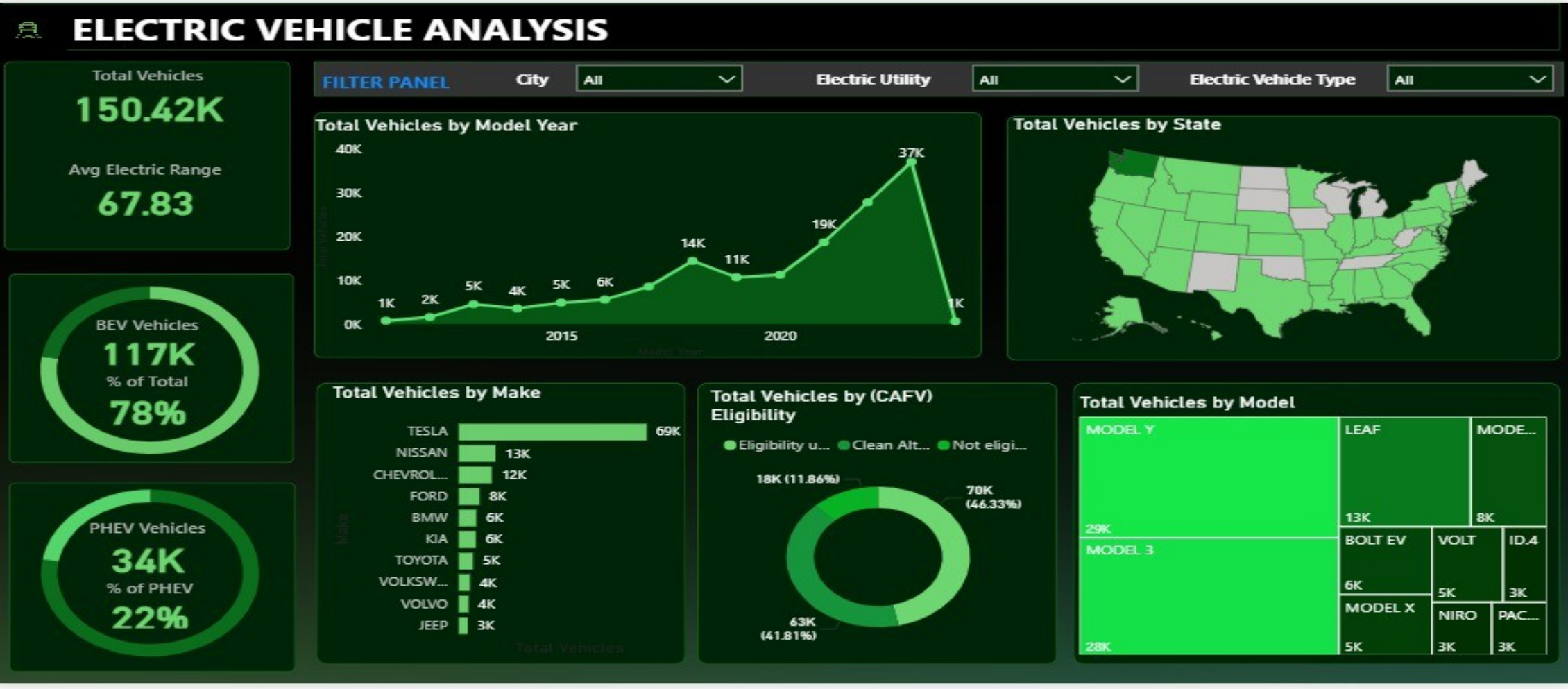
Treemap

Top 10 by Model

Consumer preference across specific EV models

Tools: SQL (data modeling) • Power BI (visualization) • DAX (KPI calculations)

Interactive Power BI Dashboard



KEY INSIGHTS

What the Data Reveals

150.42K

Total EVs analyzed across the U.S.

78%

of vehicles are Battery Electric (BEV) vs. 22% PHEV

46%

Tesla market share, led by Model Y & Model 3

42%

of vehicles are confirmed CAFV-eligible

Growth Trajectory

Adoption grew from **under 5K vehicles in 2015** to a peak of **37K in 2023** — a clear signal of accelerating market adoption and readiness for continued investment in EV infrastructure.

Manufacturer Landscape

Tesla dominates with **69K vehicles**, followed by Nissan (13K) and Chevrolet (12K) — indicating a concentrated but growing competitive field.

CONCLUSION

Business Impact & Takeaways

Investors & manufacturers can prioritize regions and models showing the fastest adoption growth.

Policymakers can use CAFV eligibility gaps (58% not yet confirmed) to refine incentive programs.

Charging infrastructure planning can align with state-level EV density revealed by the map view.

The dashboard's drill-down filters (City, Utility, Vehicle Type) enable self-serve exploration for stakeholders.

Thank You

Let's connect to discuss this project further

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